





F4SIGHT

Every border post already has eyes. Most are never watched.

DRISHTI-BOP turns the CCTV already installed at a Border Out Post into a live intrusion-detection network — detection, tracking, scored alerts, and cryptographically verifiable evidence. Built for zero marginal cost, entirely offline.

EVENT

Smart India Hackathon 2026

PROBLEM STATEMENT

26187

THEME

Blockchain & Cybersecurity

TEAM

SW-73 · ByteForge

● 01 · THE FAILURE MODE — WHAT WE'RE SOLVING

A camera doesn't fail loudly. A tired operator does.

A Border Out Post already records everything. But recording is not watching — and the failure mode of a human staring at a wall of feeds for an 8-hour shift is **silent**: nobody can say which frames were missed, or when attention lapsed.

THE GAP

Feeds exist, review doesn't scale past a handful of screens per operator.

THE COST

Every new camera or sensor today means a licence fee, a subscription, or a cloud bill.

THE TRUST PROBLEM

Even a caught incident is only as good as the evidence chain behind it in a courtroom.

● 02 · THE APPROACH — WHAT F4SIGHT DOES

Bolt analytics onto the RTSP streams that already exist.

No new hardware, no cloud account, **₹0 marginal cost**. F4SIGHT detects, tracks and reasons about what a camera sees; scores every candidate event through an additive, fully explainable model; raises debounced alerts with real evidence; and anchors that evidence to a tamper-evident ledger.

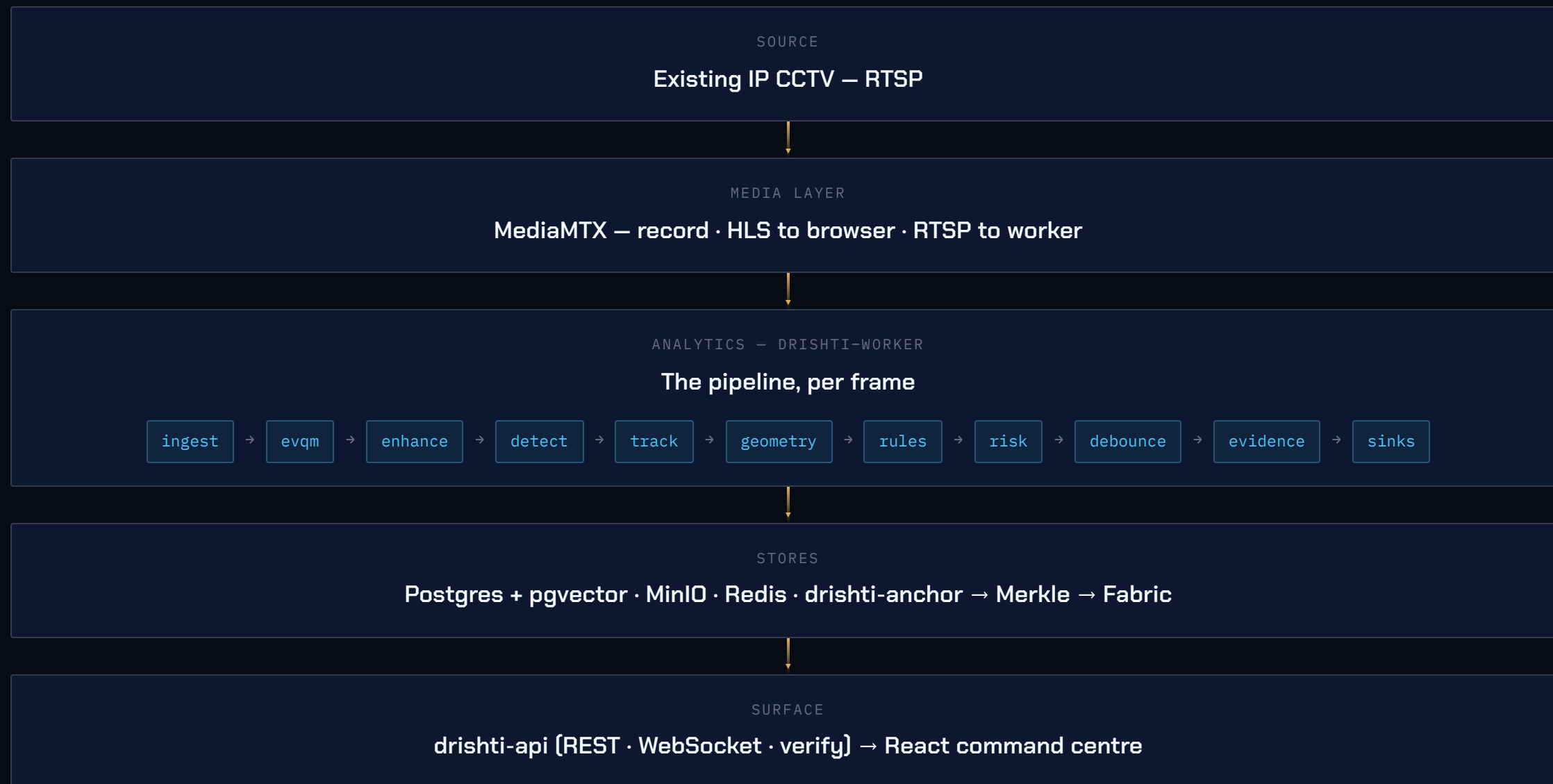
WHAT IT DOES

Intrusion zones & tripwires · person/vehicle tracking · ANPR with HMAC'd plates, never plaintext · explainable risk scoring · environmental self-adaptation (EVQM) · an operator console built for keyboard-first triage.

WHAT IT DELIBERATELY DOES **NOT** DO

No automated response — no barrier, siren or dispatch, ever. No biometric enrolment — faces match only an explicit watchlist. No plaintext plate storage — a DB leak is never a list of who drove where.

One RTSP stream in. A verifiable alert out.



THREADED, NOT ASYNC

The hot path is CPU/GPU-bound and every heavy call releases the GIL — one process, one shared accelerator.

DROPS THE OLDEST FRAME

On a slow host, analytics run on **recent** frames.
An alert about where someone was 30s ago isn't an alert.

ANCHORING NEVER BLOCKS ALERTING

`ledger_status='pending'` is a perfectly good alert.

The evidence is hashed before anyone can argue with it.

Every alert's original frame, metadata and geometry are canonicalised under **RFC**

8785, hashed with **SHA-256**, batched into a **Merkle tree**, and the root anchored to a permissioned ledger. Tamper one byte later, and the verification panel says so — by name.

```
$ python scripts/walkthrough.py
```

```
— Risk: additive and explainable —
```

```
TRIPWIRE_CROSS          +45.00
```

```
NIGHT_MOVEMENT          +20.00
```

```
SCORE                    65.00  severity=high
```

```
contributions sum to score: True
```

1→257

Merkle tree sizes verified by property-based test — forged leaves never pass

400+

Generated signal combinations checked: risk contributions always sum to the score

0

Plaintext number plates ever stored — HMAC only

Nine dependencies this repo couldn't hand itself. Six closed, live.

Every external dependency a demo can quietly skip — hardware, footage, a GPU, a real camera — named honestly, with a working fallback either way.

GATE 1

CLOSED

Docker stack

Postgres, MinIO, Redis, MediaMTX — run together live; two upstream breaks found & fixed.

GATE 2

CLOSED

Model weights

YOLO11n/s exported to ONNX and verified on two independent machines.

GATE 3

SMALL SAMPLE

Labelled footage

MOT16 clips, honest metrics published — not a production accuracy claim.

GATE 4

CLOSED

Real RTSP camera

A live phone camera, tracked and alerting — orientation bug found & documented.

GATE 5

CLOSED

GPU acceleration

CUDA measured on an RTX 3050 — three silent perf bugs found and fixed along the way.

GATE 6

OPEN

Hyperledger Fabric

Mock hash-chained ledger fallback — same code path, already caught real corruption.

GATE 7

POLICY

Deployment decisions

Licence path, face analytics, site geometry — deliberately left to the deploying org.

GATE 8

CLOSED

Visual/UX verification

Screenshotted and debugged live — two real UI bugs caught this way, fixed.

GATE 9

SCOPED

Face matching (\$7.10)

Disabled by default by design — everything around it is built, module itself is Phase 10.

• 06 • PERFORMANCE — MEASURED, NOT ESTIMATED

A laptop GPU a tier below spec, already ahead of the \$5 target.

Measured live on an RTX 3050 Laptop (4GB VRAM) — one tier below the RTX 3060

this profile targets.

40.1_{ms}

CPU, ORT CPU EP — YOLO11s @ 640×640,
batch 1

15.2_{ms}

CUDA EP, same model & batch — **2.6× faster**

76_{fps}

CUDA EP aggregate throughput at batch 8 —
12.9ms/frame

6.4_{cams}

Concurrent cameras sustained at the
profile's 12fps target

VERIFIED AGAIN DURING THIS VERY BUILD — YOLO11N @ 480×480, RTX 3050

CPU **34.3ms**/frame → CUDA **14.5ms**/frame · **2.4× faster, reproduced live**

Free and open at every layer. No metered inference, anywhere.

LAYER	CHOICE	LICENCE
Detection	YOLO11n/s ONNX · RT-DETR-R18 fallback	AGPL-3.0 · Apache-2.0
Tracking	ByteTrack, reimplemented in-repo	MIT (algorithm)
ANPR	YOLO11n plate + PaddleOCR v4	AGPL-3.0 · Apache-2.0
Faces (opt-in)	InsightFace SCRFD + ArcFace	MIT
Inference	ONNX Runtime (+ TensorRT / CoreML)	MIT
Database	PostgreSQL 16 + pgvector	PostgreSQL Licence
Media	MediaMTX	MIT
Ledger	Hyperledger Fabric 2.5	Apache-2.0
API / Dashboard	FastAPI · React + Vite + Tailwind	MIT

• 08 • GETTING IT RUNNING — THE WHOLE DEMO, ONE COMMAND

Bare clone to running app. One command, ₹0.

```
$ git clone <repo> && cd f4sight
$ make demo
1/6 migrations      alembic upgrade head
2/6 seed            demo site, cameras, zones, watchlist
3/6 models          ~166MB, skipped after first run
4/6 fixtures        sample RTSP feeds
5/6 dashboard build tsc -b && vite build
6/6 start           worker + api + anchor, together
ready → http://localhost:8000/
```

EVERY STEP ALONE

make **install** · **up** · **migrate** · **seed** · **models** · **fixtures** — each idempotent, each safe to re-run.

PROVE IT

make **test** — 384 Python + 6 TS tests. make **bench** — writes docs/BENCH.md. make **eval** — metrics table.

BRING YOUR OWN CAMERA

Any empty Live Wall slot: click, type an IP. Live within seconds — the detector is already loaded and shared.



Vigilance at the border, powered by sight.

F4SIGHT — DRISHTI-BOP. Built end to end, free, for Smart India Hackathon 2026.

PS 26187

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